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Ministry of Higher Education
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وزارة التعليم العالي
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AI342 Course Project-F03

Advanced Digital Image Processing (DIP)

Supervised by Dr. Nouf Alharbi

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Table of Contents

Table of Figures

Abstract

In this project, we created an image processing system that includes different stages, starting with noise injection into images or dealing with images containing preexisting noise and identifying the source of distortion.

Then, we applied the appropriate filter to process different types of noise, such as salt and paper noise.

We also applied segmentation and morphology to identify different shapes in the images and extract their distinctive features, which were used to classify the images and assign them appropriate labels.

The images were analyzed, and samples were tested in test data against training data sets. Different images were compared, and the appropriate analysis was written for each in terms of noise types and suitable filters.

1. Introduction

Image processing is a crucial field in computer vision and artificial intelligence, enabling computers to automatically analyze, enhance, and understand digital images. Its primary goals are to improve image quality, extract information, and prepare images for further analysis and classification.

In this project, various image processing techniques were applied to images of two different types of fruit to study enhancement, segmentation, morphology, and classification.

Different noise patterns were added to the images, such as salt and pepper noise, Gaussian noise, periodic noise, Rayleigh noise, and regular noise. Several spatial and frequency filters were applied to determine the optimal recovery method for each noise type.

Image segmentation and morphological processing, such as global thresholding, adaptive thresholding, and Otsu thresholding, were applied to separate the foliage from the background. A morphological process, including opening, closing, dilation, erosion and border extraction, was then applied to enhance image elements, remove noise, and preserve important structural details.

Feature extraction includes several types: Area, Perimeter, Circularity, Compactness ،Hu_1 and Solidity

Finally, the features extracted from the segmented images were analyzed and compared to classifying the images according to their geometric properties.

1.1 Dataset Overview

The dataset used in this project was obtained from a fruit dataset available on Kaggle. This dataset contains high-quality images of nine different types of fruit. From the original dataset, we selected a subset of 30 images from two different categories: apples and bananas. The reason for choosing fruit images is that they offer clear geometric variation in shape, size, edge structure, and elongation, making them suitable for evaluating image restoration techniques and object description.

2.Individual Analysis

In our initial attempts to classify the bananas and apples into two groups, we encountered a problem: the classifier was handling the data into a new, unknown class. After investigation, it became clear that the variation in apples in the images stemmed from the presence of images of apples with the fruit leaf and images without the leaf.

This affected the classification of the data samples because segmentation is very sensitive to object separation and property extraction. Our use of features includes area, perimeter, and circularity, and these properties are highly sensitive to classification and property extraction. Therefore, we added a new class to the classification: "Apples with Leaf." Thus, the classification is based on "Bananas," "Apples," and "Apples with Leaf", which is the optimal solution for handling variations in fruit shape.

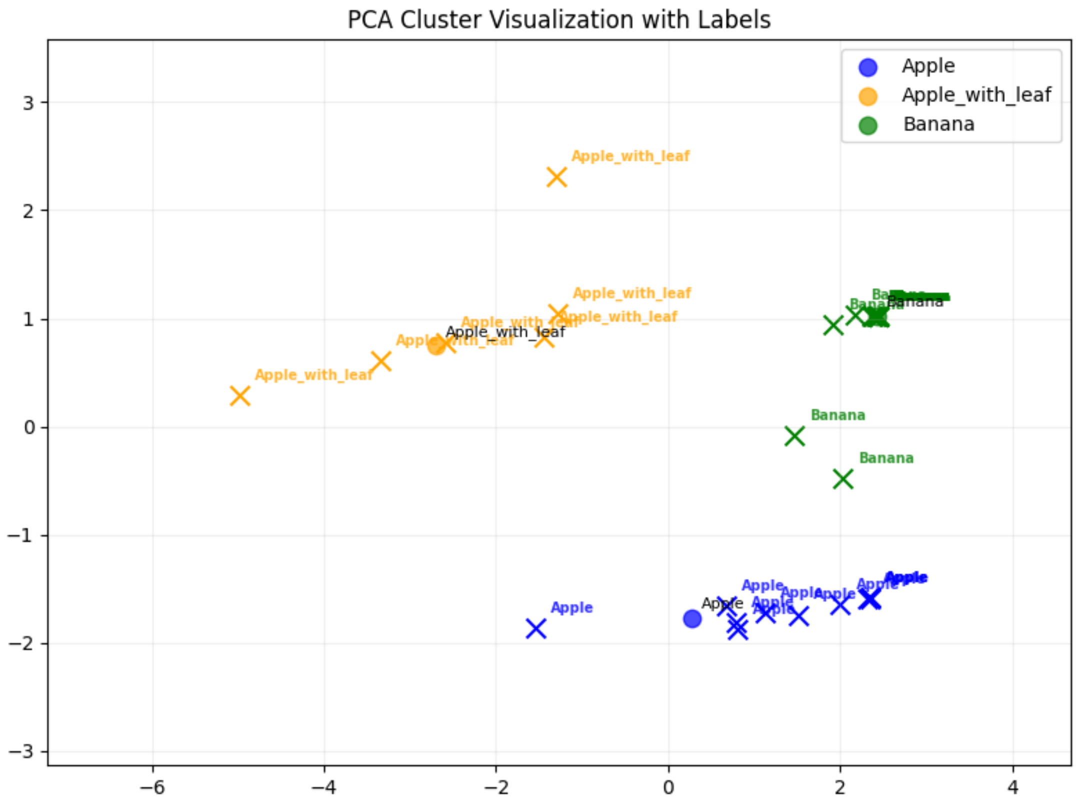


Figure 1:PCA Cluster Visualization

The preprocessing and feature extraction pipeline was adapted from the course laboratory exercises [1].

Area	Perimeter	Circularity	Compactness	Hu_1	Solidity	Class	Prediction
121801.0	1396.0	0.785	16.0	0.166666666666667	1.0		Banana
121801.0	1396.0	0.785	16.0	0.1666666666666666	1.0		Banana
56.0	39.8	0.444	28.285	0.2279118771764434	0.704		Apple
346.0	156.4	0.178	70.667	0.5521185751633558	0.486		Apple
152.0	64.8	0.455	27.599	0.3304280555454583	0.852		Apple
181.0	61.9	0.593	21.197	0.2015352825322499	0.866		Apple
55511.5	2556.8	0.107	117.759	0.527010095905861	0.456		Apple_with_leaf
121801.0	1396.0	0.785	16.0	0.1666666666666666	1.0		Banana
121534.5	1430.2	0.747	16.83	0.1668828753863068	0.998		Banana
61121.0	1248.8	0.493	25.515	0.1682414727539174	0.923		Banana
121801.0	1396.0	0.785	16.0	0.1666666666666666	1.0		Banana
121783.5	1410.9	0.769	16.345	0.1666645080314963	1.0		Banana
221.5	56.9	0.861	14.601	0.1606714185394964	0.959		Apple
963.0	167.9	0.429	29.267	0.1941136116738845	0.851		Apple
430.0	85.1	0.746	16.847	0.1982868552454351	0.952		Apple
176.0	50.6	0.863	14.563	0.1602100550963686	0.957		Apple
172.5	50.0	0.866	14.517	0.1605266353264811	0.956		Apple
203.5	54.9	0.849	14.795	0.1618486145528619	0.953		Apple
2696.5	306.5	0.361	34.837	0.2550469907193979	0.823		Apple
74377.0	2586.0	0.14	89.914	0.338678263178029	0.611		Apple_with_leaf
121801.0	1396.0	0.785	16.0	0.1666666666666666	1.0		Banana
121801.0	1396.0	0.785	16.0	0.1666666666666666	1.0		Banana
47502.5	2504.5	0.095	132.049	0.6721738311700238	0.39		Apple_with_leaf
121578.5	1522.3	0.659	19.06	0.1667074229856786	0.998		Banana
67244.0	2439.4	0.142	88.494	0.3618539218293234	0.552		Apple_with_leaf
121801.0	1396.0	0.785	16.0	0.1666666666666666	1.0		Banana
44295.0	882.2	0.715	17.57	0.1781937926136418	0.975		Banana
88546.5	3692.3	0.082	153.963	0.1698724232682591	0.852		Apple_with_leaf
121801.0	1396.0	0.785	16.0	0.1666666666666666	1.0		Banana
121801.0	1396.0	0.785	16.0	0.1666666666666666	1.0		Banana
121801.0	1396.0	0.785	16.0	0.1666666666666666	1.0		Banana
115163.5	1575.4	0.583	21.552	0.1664216238495908	0.97		Banana
121744.0	1422.5	0.756	16.621	0.1666887433904355	1.0		Banana
121801.0	1396.0	0.785	16.0	0.166666666666667	1.0		Banana
121801.0	1396.0	0.785	16.0	0.166666666666667	1.0		Banana
121801.0	1396.0	0.785	16.0	0.166666666666667	1.0		Banana
34958.0	2389.6	0.077	163.349	1.02669393028665	0.287		Apple_with_leaf

Figure 2 :Prediction Table

Each person in the group is responsible for 5 images of the Restoration, Enhancement, Segmentation and Morphology application, and for analyzing the results and parameters used.

Note: Two attempts for each operation were recorded for each image in the table.

The following tables represent the results of **Shumukh's** experiments on 5 images:

Phase 1: Restoration & Enhancement

Image	Noise Type	Intensity	Transformation	Filter	Kernal Size	PSNR	MSE
CA-1	Salt & Pepper	0.07	Gamma (0.97)	median	5	32.14	39.70
			Log	mean	11	21.50	460.12
CA-2	Gaussian	0.075	None	Median	5	32.75	34.49
			Log	Median	7	27.72	109.87
CB-1	Periodic	0.055	log	Notch (D = 20)	3	22.07	403.91
			Gamma (1.2)	Low pass (D=20)	3	20.03	645.43
CB-2	Rayleigh	0.06	Gamma (1.2)	median	3	32.02	40.82
			none	mean	3	27.11	126.55
CB-3	Uniform	0.065	Gamma (1.1)	median	3	35.76	17.28
			log	mean	5	20.32	603.62

Phase 2: Segmentation & Morphology

Image	Operation	Method	Threshold	Shape	Size
CA-1	Closing	Otsu	-	Disk	5
CA-2	Closing	Otsu	-	disk	5
CB-1	Erosion	Global	200	disk	15
CB-2	opening	adaptive	-	disk	13
CB-3	closing	global	150	disk	3

Final analysis:

The results showed that choosing the correct filter based on the noise type improved image quality significantly. Median filtering gave the best results in most images, especially when combined with Gamma transformation. The best overall result was obtained for P3-CB with Uniform noise, achieving the highest PSNR and lowest MSE values, which indicates very

good restoration quality. Gaussian and Salt & Pepper noise were also removed effectively using Median filtering. On the other hand, Mean filtering produced weaker results with higher error values. Periodic noise in P1-CB was the most difficult case because frequency noise was harder to remove completely even when using a Notch filter.
In the segmentation phase, Closing operations with Disk structuring elements produced the best object shapes and smoother boundaries. Adaptive thresholding worked better for images with uneven lighting, while Global thresholding was effective for simpler cases. Larger structuring element sizes helped remove unwanted regions in difficult backgrounds. Overall, the experiments showed that selecting suitable restoration and morphology techniques is important for obtaining accurate segmentation results.

The following tables represent the results of **Bushra's** experiments on 5 images :

Phase 1: Restoration & Enhancement

Image	NoiseType	Intensity	Transformation	Filter	Kernal Size	PSNR	MSE
CA-7	Rayleigh	0.05	Gamma (1.17)	Median	3	36.30	14.24
			Gamma (1.17)	Mean	7	35.97	16.46
CA-8	Periodic	0.05	None	Low Pass (D0 = 195)	9	29.29	76.53
			None	Notch (D0 = 84)	5	29.04	81.19
CA-9	Gaussian	0.11	None	Median	9	02.30	69.64
			None	Mean	5	27.17	124.62
CB-10	Uniform	0.07	Gamma (1.2)	Mean	3	34.41	23.57
			None	Median	3	34.11	25.25
CB-11	S&P	0.12	None	Median	5	38.02	10.26
			None	Low Pass (D0 = 32)	5	21.37	444.64

Phase 2: Segmentation & Morphology

Image	Operation	Method	Threshold	Shape	Size
CA-7	Opening	Global	154	Square	5
CA-8	Erosion	Global	91	Square	15
CA-9	Erosion	Otsu	210	Square	5
CB-10	Erosion	Global	215	Disk	9
CB-11	Opening	Global	227	Square	11

Final analysis:

The Median filter proved to be the most effective restoration tool through the samples, especially for Rayleigh, gaussian and salt & pepper noise, with lowest MSE 10.26 and highest PSNR 38.02 for salt and pepper noise, and similar results with gaussian noise, without any need of intensity transformation, showing high ability of removing noise and extreme pixel outliers while preserving edge structure.
combining Gamma transformation (1.17) with median filter, performed best for Rayleigh noise, because gamma adjusted the brightness distribution before filtering, which enhanced the restoration outcome. in the case of periodic noise a frequency domain approach was necessary. The Low Pass filter with D0=195 showed best result. The large D0 value preserved sufficient image detail.

The following tables represent the results of **Razan's** experiments on 5 images :

Phase 1: Restoration & Enhancement

Image	Noise Type	Intensity	Transformation	Filter	Kernal Size	PSNR	MSE
CA-10	Salt & P	0.1	None	Median	5	28.8	85.76
			None	Mean	7	23.82	269.62
CA-11	Gaussian	0.05	None	Median	5	35.61	17.88
			None	Mean	5	33.92	33.2
CA-12	Periodic	0.04	Log	Low Pass D0=200	-	30.54	57.37
			Gamma (0.66)	Median	3	21.04	511.87
CB-12	Rayleigh	0.60	Gamma (1.2)	Mean	3	29.99	65.23
			Log	Notch D0=10	-	22.98	327.14
CB-13	Uniform	0.11	Gamma (1.2)	Mean	3	34.28	24.28
			Log	Notch D0=10	-	18.95	828.3

Phase 2: Segmentation & Morphology

Image	Operation	Method	Threshold	Shape	Size
CA-10	Erosion	Global	205	Disk	15
CA-11	Closing	Otsu	26	Disk	9
CA-12	Closing	Global	225	Disk	15
CB-12	Dilation	Adaptive	-	Disk	3
CB-13	Dilation	Otsu	129	Disk	7

Final analysis:

The results showed that the Median filter performed better for Salt & Pepper and Gaussian noise because it preserved edges while reducing noise. Frequency-domain filters such as Low Pass filtering performed well for periodic noise, especially when suitable D0 values were selected. Mean filtering smoothed random intensity variations; therefore, it was more suitable for Rayleigh and Uniform noise.
Gamma and Log transformations also affected image contrast and brightness. The best restoration results were identified by achieving higher PSNR values and lower MSE values. Segmentation and morphology operations were applied to enhance object extraction from the images. Global and Otsu thresholding performed well when object and background intensities were clearly separable. While Adaptive thresholding was more suitable for images with uneven illumination. Erosion was used to remove small noisy components, Dilation expanded object regions, and Closing filled small gaps inside the segmented objects. Disk-shaped structuring elements were appropriate for preserving object details and producing smoother boundaries.

The following tables represent the results of **Layal's** experiments on 5 images :

Phase 1: Restoration & Enhancement

Image	Noise Type	Intensity	Transformation	Filter	Kernal Size	PSNR	MSE
CA-5	Salt & pepper	0.05	None	Medien	3	31.15	49.93
			Gamma (1.03)	Laplacian	7	13.71	2050.76
CA-6	Gaussian	0.096	None	Mean	3	27.08	127.37
			Gamma (0.92)	Low Pass d0(50)	7	26.44	147.56
CB-9	periodic	0.1	Gamma (0.99)	Medine	15	28	102.98
			Gamma (0.79)	Noth d0(71)	3	24.44	233.96
CB-8	Rayleigh	0.075	Gamma (1.33)	Mean	3	29.97	112.86
			Gamma (1.51)	Noth d0(50)	7	26.34	150.76
CB-7	uniform	0.17	Gamma (1.62)	Mean	5	27.61	112.35
			Log	Low Pass d0(73)	3	25.44	185

Phase 2: Segmentation & Morphology

Image	Operation	Method	Threshold	Shape	Size
CA-5	Erosion	Global	150	Sequere	3
CA-6	Dilation	Otsu	-	Cross	3
CB-7	Erosion	Global	195	Sequere	9
CB-8	Closing	Global	229	Sequare	3
CB-9	Closing	Adaptive	150	Sequare	3

Final analysis:

The Median filter is considered a powerful and flexible tool for noise processing, with its method of use varying depending on the situation. In the case of Salt & Pepper noise, this filter has proven to be the optimal choice effectively and without the need for any transformations; due to its high efficiency in removing extreme white and black spots without distorting the features and edges of the image. The same filter was used to improve Periodic noise, but with different settings; the filter was applied with an exceptionally large kernel size of up to 15, in conjunction with using a Gamma transform with a factor of 0.97 to adjust the image brightness in a way that ensures the best possible restoration of its details.

The mean filter stands out as an effective and versatile tool, having proven successful in handling three different types of noise due to its flexibility and the ability to integrate it with other transforms. In the case of Gaussian noise, using the mean filter alone was the most successful option, as this simple filter was able to efficiently smooth the random variations of

Gaussian noise. The same filter also demonstrated excellent efficiency when dealing with noise that required additional adjustment; in Rayleigh noise, the image was successfully restored by applying the mean filter combined with contrast adjustment using a gamma transform]. In a very similar manner, this combination proved successful in handling uniform noise, where the mean filter was successfully applied in conjunction with a gamma transform, confirming the strength of this filter when supported by the appropriate mathematical transform.

The following tables represent the results of **Hatoon's** experiments on 5 images :

Phase 1: Restoration & Enhancement

Image	Noise Type	Intensity	Transformation	Filter	Kernal Size	PSNER	MSE
CA-3	Uniform	0.6	Gamma(1.19)	Low pass (D0:121)	5	31.42	46.90
			None	Laplacian	5	17.77	1086.17
CA-4	Salt& pepper	0.063	None	Median	3	37.31	12.08
			None	Low pass (D0:39)	3	25.13	199.45
CB-4	Gaussian	0.07	None	Mean	3	30.29	60.80
			Gamma(1)	Low pass (D0:85)	5	27.67	111.22
CB-5	Reyleigh	0.08	Gamma(1.19)	Low pass (D0:82)	3	24.83	213.98
			Log	Notch (D0:90)	3	20.6	563.35
CB-6	Periodic	0.05	None	Mean	3	30.05	64.25
			None	Notch (D0:85)	3	29.05	80.90

Phase 2: Segmentation & Morphology

Image	Operation	Method	Threshold	Shape	Size
CA=3	Boundary Extraction	Otsu	180	Cross	3
CA-4	Closing	Otsu	238	Disk	3
CB-4	Opening	Otsu	255	Square	11
CB-5	Dilation	Adaptive	255	Cross	3
CB-6	Boundary Extraction	Global	230	Square	15

Final analysis:

In the case of **Salt & Pepper noise**, the **Median Filter** achieved the best performance with the highest PSNR value (37.31) due to its ability to remove impulsive noise effectively by selecting median pixel values. In contrast, the **Low Pass Filter** (D0 = 39) showed weaker results because it only smoothed the noise rather than removing it directly.
For image transformations, the **Gamma Transformation** (1.19) performed well combined with Low Pass Filters for **Uniform** and **Rayleigh noise**, improving contrast and increasing PSNR values to (31.42) and (24.83), respectively. However, Log **Transformation** produced poor results with the **Notch Filter** because it amplified noise in dark regions.
Additionally, the **Mean Filter** with a (3×3) kernel showed stable performance for Gaussian and periodic noise, especially with low noise intensity (0.05). On the other hand, the **Laplacian Filter** achieved the weakest performance with Uniform noise, recording the lowest PSNR (17.77) and highest error rate, since it enhanced noise instead of suppressing it.
Overall, the results indicate that kernel size and transformation type must be carefully selected according to the filter and noise characteristics. Larger kernels (5×5) were more effective for strong smoothing, while smaller kernels preserved edges and fine details more efficiently.

The following tables represent the results of **Shatha's** experiments on 5 images :

Phase 1: Restoration & Enhancement

Image	Noise Type	Intensity	Transformation	Filter	Kernal Size	PSNR	MSE
CA-13	periodic	0.05	None	Notch	5	31.87	42.29
			Gamma(0.62)	notch	5	23.40	297.27
CA-14	Rayleigh	0.096	Gamma(1.48)	Mean	5	26.78	136.63
			None	Mean	5	21.21	491.66
CA-15	uniform	0.07	None	Laplacian	5	27.33	120.28
			Negative(1.48)	Laplacian	7	2.13	39860.90
CB-14	S & P	0.88	LOG	Median	5	27.44	117.21
			none	mean	5	25.15	198.83
CB-15	Gaussian	0.05	None	Mean	1	46.11	1.59
			Histogram Equalization	Sobel	5	7.32	12044.31

Phase 2: Segmentation & Morphology

Image	Operation	Method	Threshold	Shape	Size
CB-14	Erosion	Otsu	127	Square	3
CB-15	Closing	Otsu	38	Cross	3
CA-13	Boundary Extraction	Adaptive	129	Disk	15
CA-14	Boundary Extraction	Global	110	Disk	15
CA-15	Dilation	Otsu	67	Disk	3

Final analysis:

The Median filter is the best to remove **Salt & Pepper** noise because it chooses the median value of the neighboring pixels so, it can withstand outlier values without blurring edges.
The Mean filter is good for most noise types, especially for **Gaussian** as it finds the average of surrounding pixels and it reduces noise efficiently if it is distributed uniformly or normally. Mean filter is not enough for **Rayleigh** noise because the noise intensity distribution is skewed. Gamma transformation was used first to correct the intensity distribution and to improve contrast before applying the Mean filter to better restoration.
For uniform noise, the **Laplacian** filter was the ideal choice without applying any transformation because it tracks changes and is very sensitive to them, as it uses the second derivation. For transformations, Log transformation was applied to the Salt & Pepper image as it was very dark. After that, filtering could be done on the low intensity regions. Gamma transformation was used for Rayleigh noise to correct the non-symmetric distribution. For the rest of the images, no transformation was needed as the illumination was balanced, and filters alone achieved satisfactory results.

3. Images and Mathematical Proof

3.1 Evaluation Metrics

To evaluate the effectiveness of the restoration filters, PSNR and MSE metrics were used. Higher PSNR values indicate better image quality, while lower MSE values indicate lower reconstruction error between the original and restored images.

$$MSE = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{y=0}^{N-1}[I(x,y) - K(x,y)]^2$$

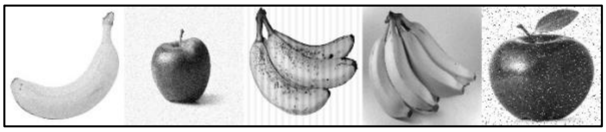
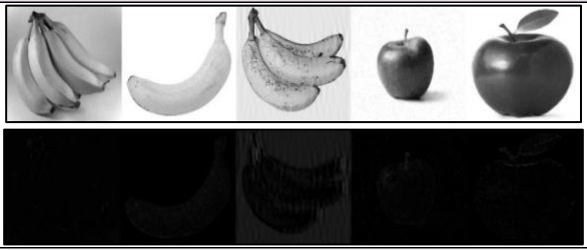
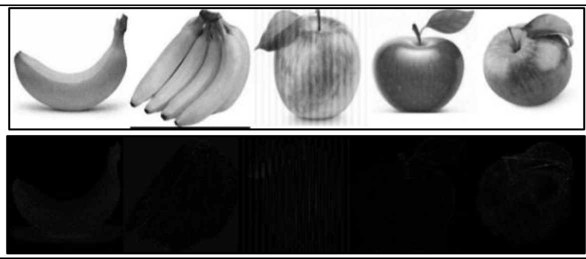
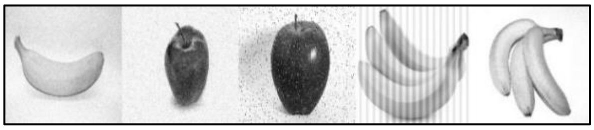
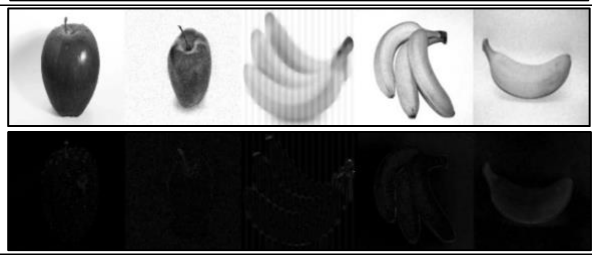
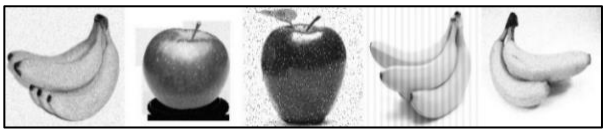
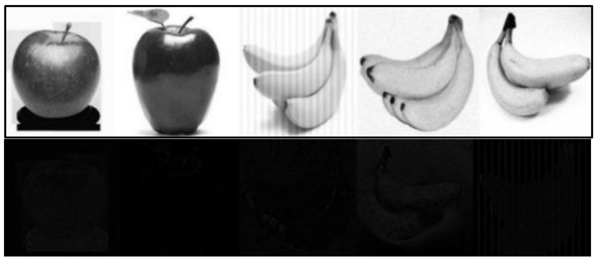
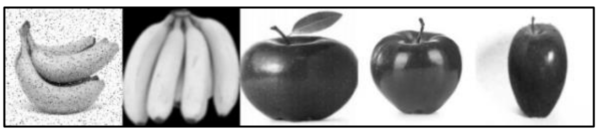
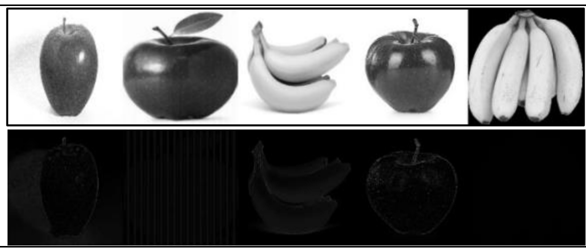
- $I(x,y)$ *represents the original image pixels value.*
- $K(x,y)$ *represents the restored image pixels value.*
- M *and* N *are the image dimensions.*

$$PSNR = 10log_{10}\left(\frac{MAX_I^2}{MSE}\right)$$

- MAX_I *is the maximum possible pixel value in the image.*
- MSE *is the Mean Sequare Error.*

3.2 Noisy Images Vs Restored Images with Diff Map

Note: The following table shows the distorted and enhanced images with Difference Map, ranked by group members according to the analysis presented above.

Noisy Images	Restored Images-Diff Map
	
	
	
	
	
	

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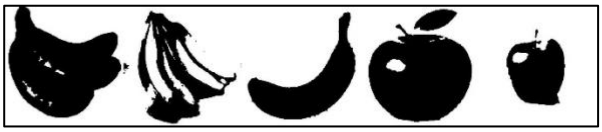
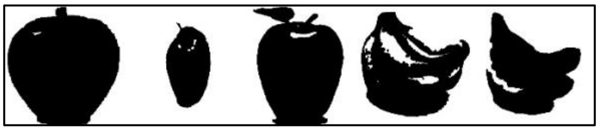
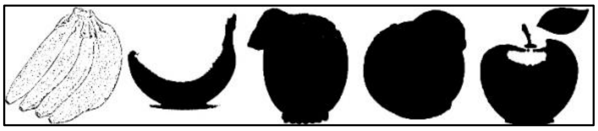
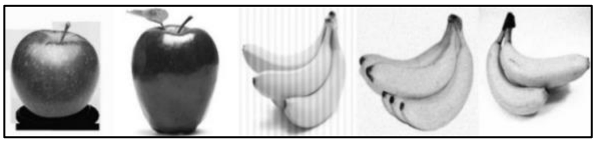
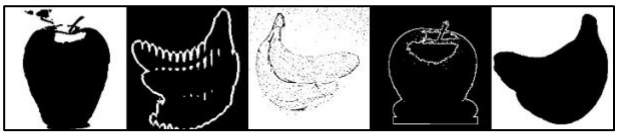
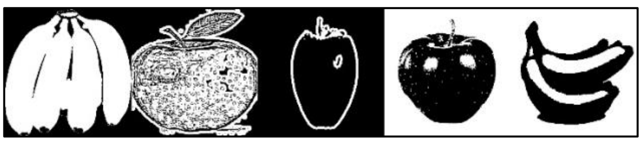
Images	Segmentation & Morphology
	
	
	
	
	
	

4.Conclusion and Key Outcomes

4.1 Group Summary Tablee

Noise Type	Best Filter	Kernal Size	PSNR	MSE
Salt & Pepper	**Median**	**5**	**38.02**	**10.26**
Gaussian	**Mean**	**1**	**46.11**	**1.59**
	Median	**5**	**35.61**	**17.88**
Periodic	**Notch**	**5**	**31.87**	**42.29**
	Low Pass (D0 = 200)	**-**	**30.54**	**57.37**
	Mean	**3**	**30.05**	**64.25**
Rayleigh	**Median**	**3**	**36.30**	**14.24**
	Mean	**3**	**29.99**	**65.23**
Uniform	**Mean**	**3**	**34.41**	**23.57**
	Low Pass (D0 = 121)	**5**	**31.42**	**46.90**

4.2 Key Outcomes

As seen in the results of the group's dataset, effective image restoration is dependent on matching the filter to the nature of the noise, and there is no universal solution towards every case.
All group's results agree that **salt & pepper noise** handled best by median filter due to its ability to eliminate extreme pixel outliers while preserving edges.
In **gaussian noise**, the noise is spread across the image following a normal distribution, approximately all pixels are affected without any pixel to be an outlier. mean filters work best because it averages all neighbours to get as similar as possible to the original image.
Periodic noise needs frequency domain filters, because it appears as repeating waves, and this is confirmed by the group's results. Low pass and notch filters do the best.
Rayleigh noise creates distributed noise with a skewed pattern. median filter handled this noise for the same reason it handles S&P noise, it preserves details while it is resistant to outliers. Lastly **uniform noise**, which affects each pixel with a fixed range, both spatial and frequency domain filters work well because there's no strict structure.
Most of the results agree that small kernel sizes (3 and 5) are best due to their ability to remove noise without blurring the image.
In conclusion, successful restoration depends on choosing the best match filter to the noise's mathematical nature and fine tuning its parameters to achieve the highest PSNR and lowest MSE possible.

References

[1] N. Alharbi, "Lab Tutorials," 2026. [Online].